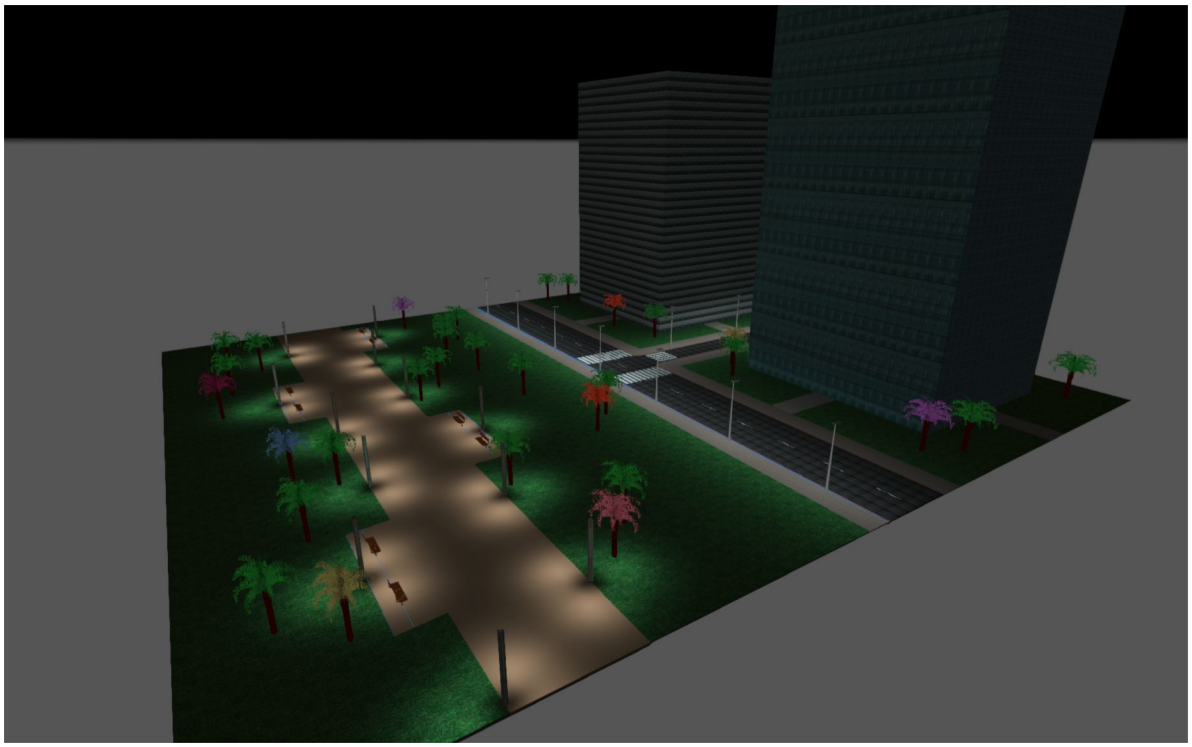
Date 4/3/2026



SITE PLAN - OUTDOOR LIGHTING DESIGN

This site plan presents the outdoor lighting design developed using DIALux software for a modern urban environment. The layout illustrates the strategic placement of luminaires across roadways, pedestrian pathways, and surrounding areas to achieve optimal illumination, safety, and visual comfort. The design has been prepared with consideration of standard lighting requirements, ensuring uniform light distribution, energy efficiency, and effective coverage of all critical zones within the site.

Created with DIALux

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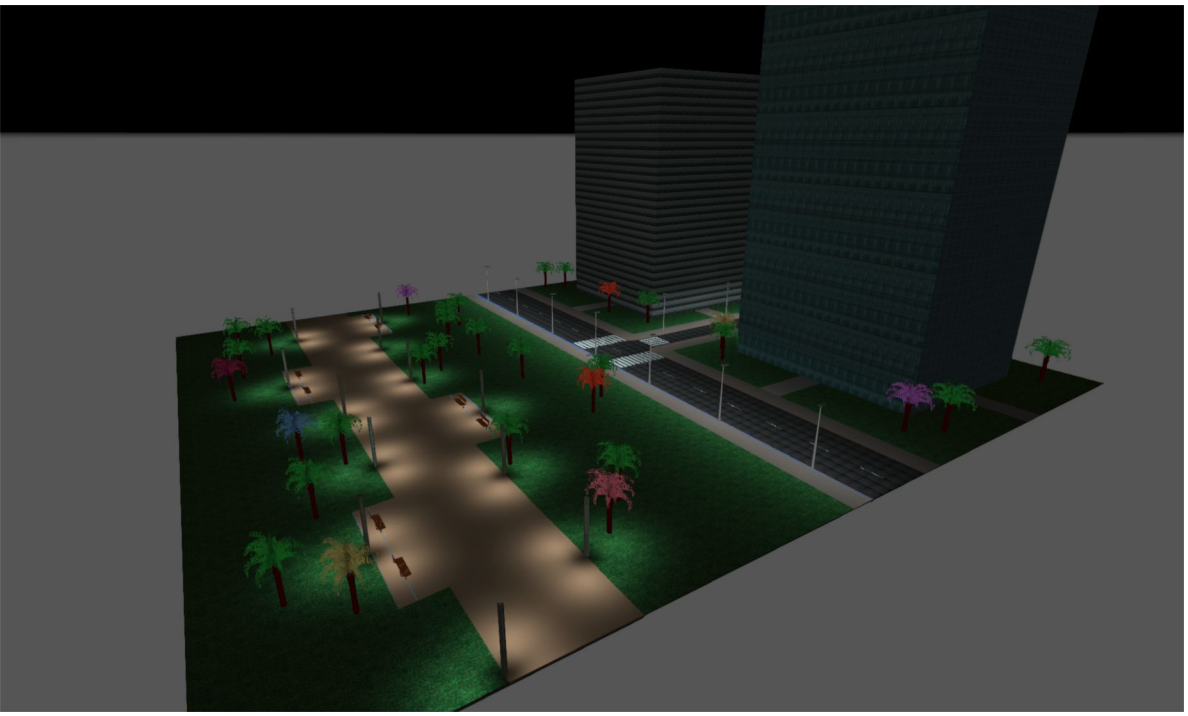
olutions



Description

This project represents an outdoor lighting design for a modern urban environment consisting of commercial buildings, pedestrian pathways, landscaped green areas, and road infrastructure. The model includes two high-rise buildings, a central roadway with pedestrian crossings, and surrounding open spaces with trees and walkways.

The primary objective of this lighting design is to ensure adequate illumination, safety, and visual comfort for both pedestrians and vehicles while maintaining energy efficiency and aesthetic appeal.

The design covers the following key areas:

Road Lighting:
Street lighting has been designed to provide uniform illumination along the roadway, ensuring safe vehicle movement and clear visibility at pedestrian crossings.
Pedestrian Pathway Lighting:
Walkways are illuminated to enhance safety and accessibility during nighttime, with attention to uniformity and glare control.

Landscape Lighting:
Green areas and trees are incorporated into the lighting plan to improve the visual environment and create an aesthetically pleasing atmosphere.

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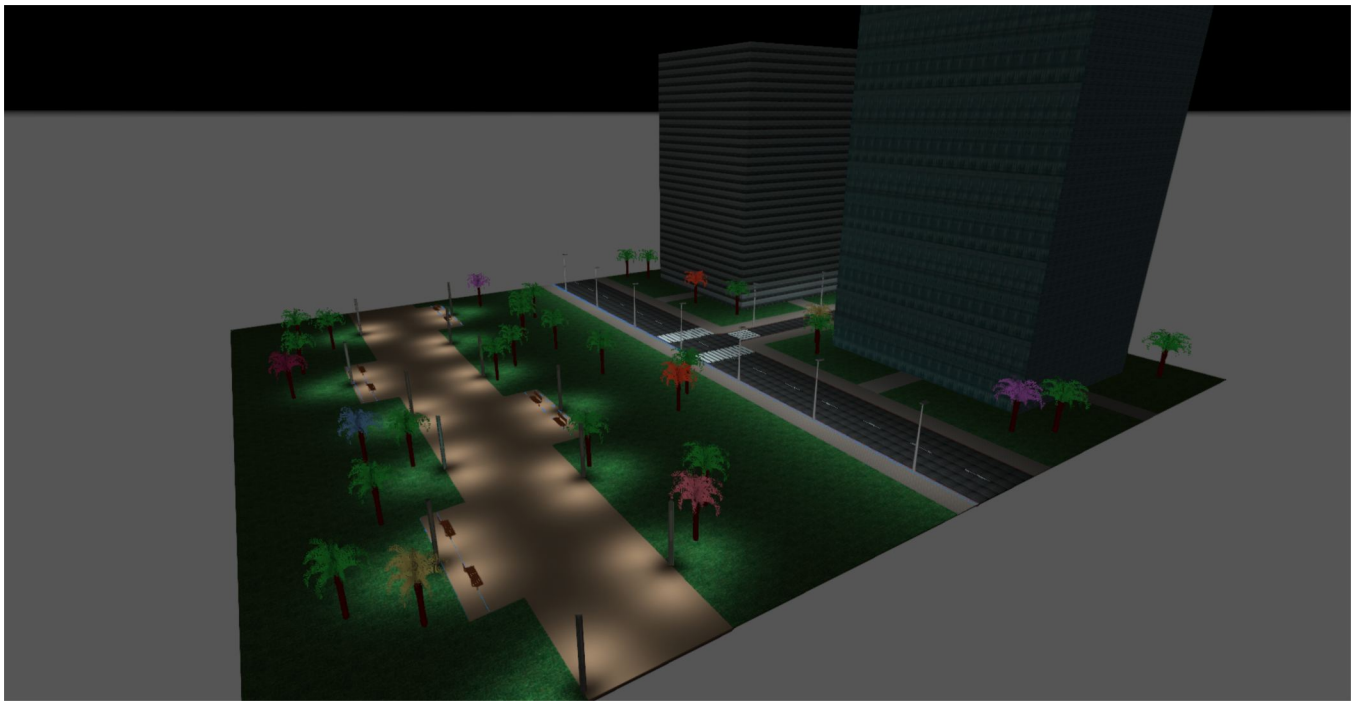
olutions

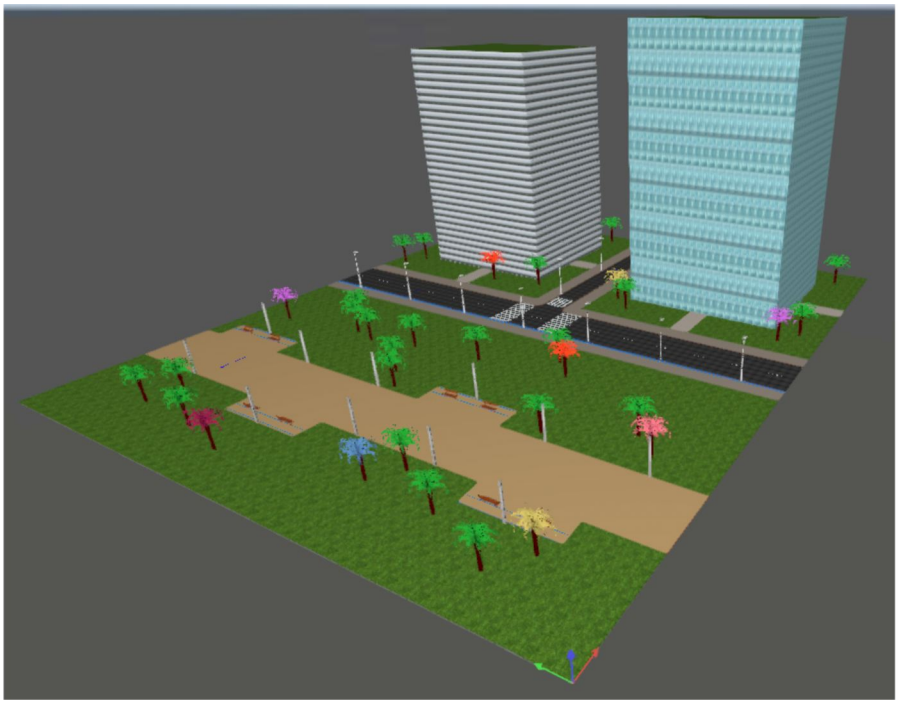
Building Surroundings:
The lighting around the buildings ensures proper visibility for entrances and surrounding areas, contributing to overall security.

The lighting calculations have been carried out using DIALux software, considering relevant standards for illuminance levels, uniformity, glare limitation, and energy efficiency.

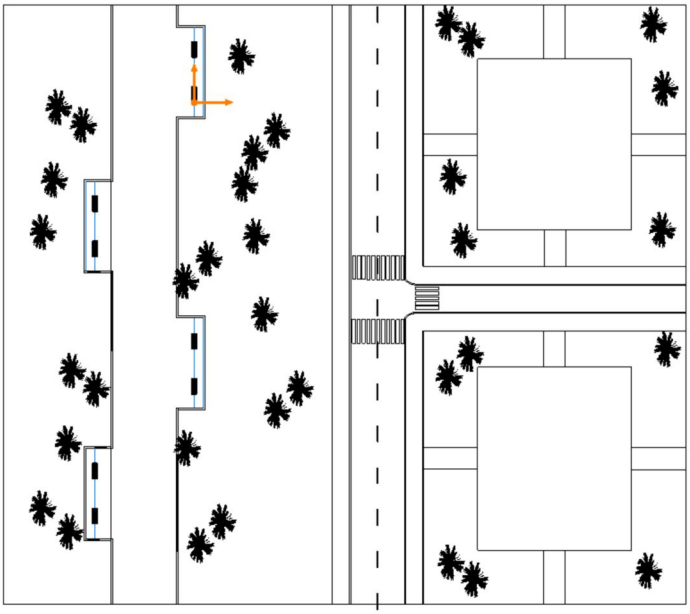
Appropriate luminaires have been selected and positioned to achieve optimal performance while minimizing light pollution and power consumption.

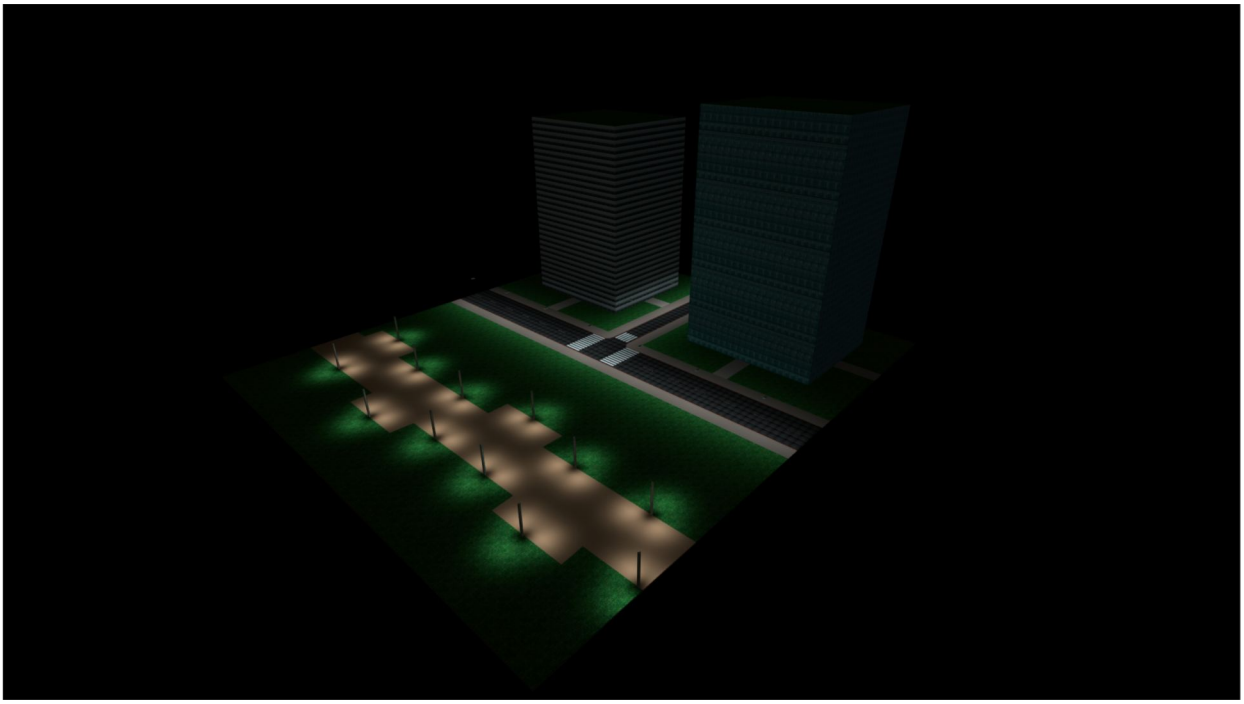
Images





Images





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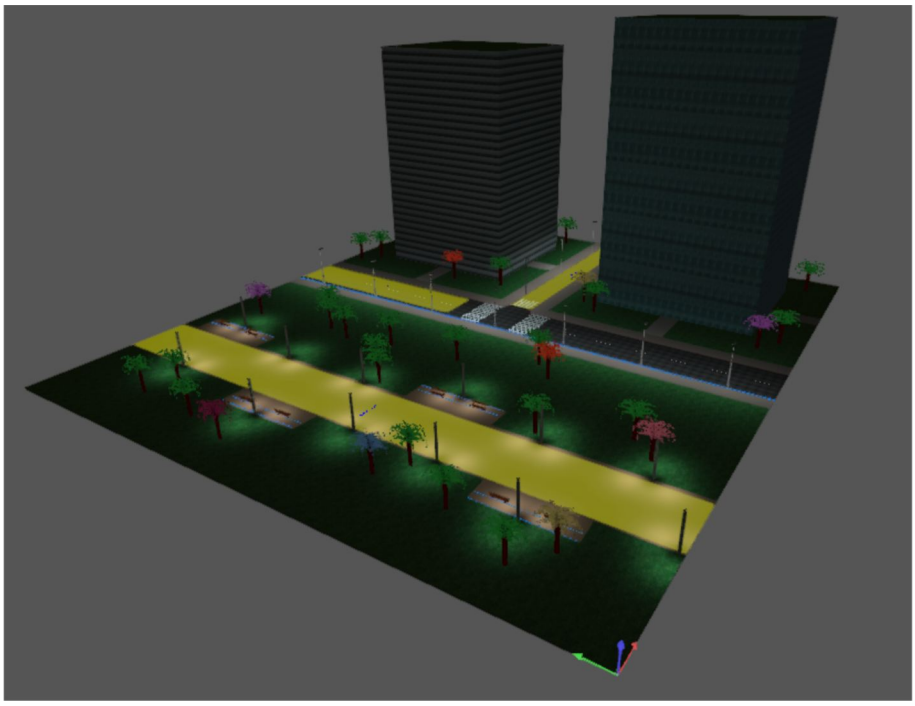
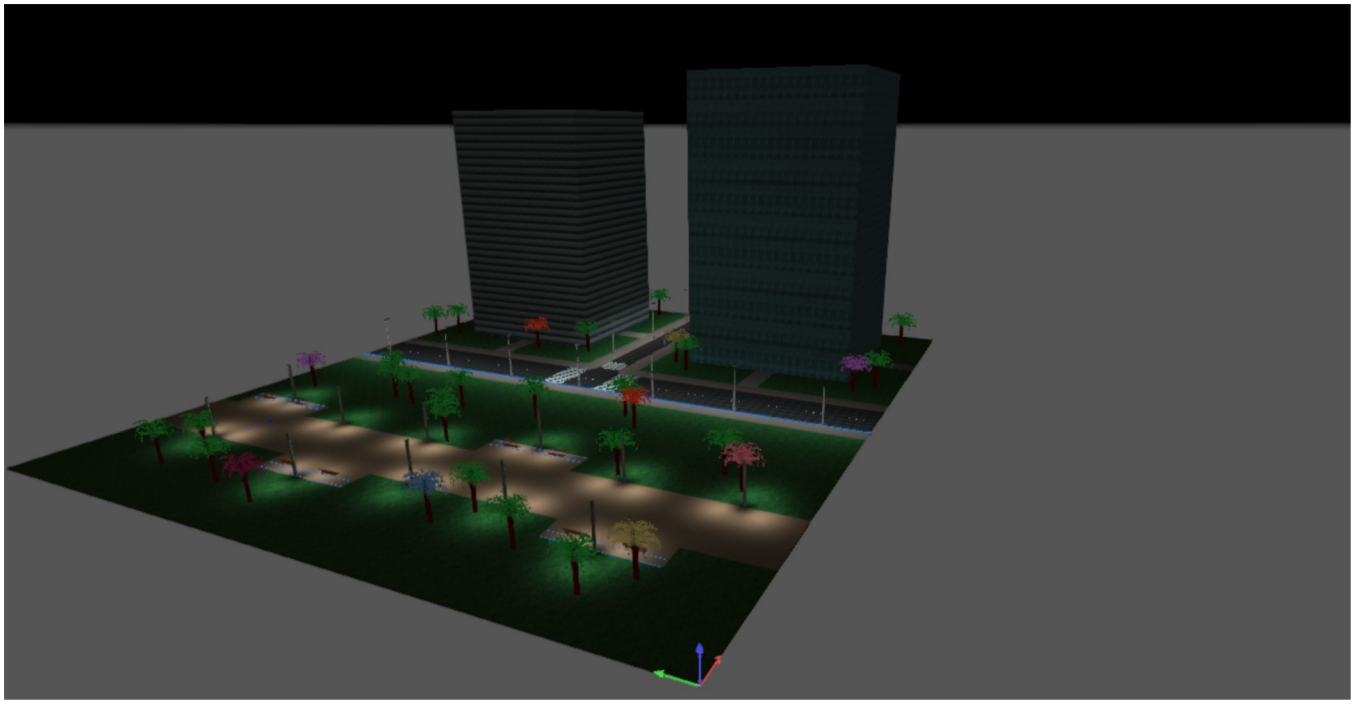
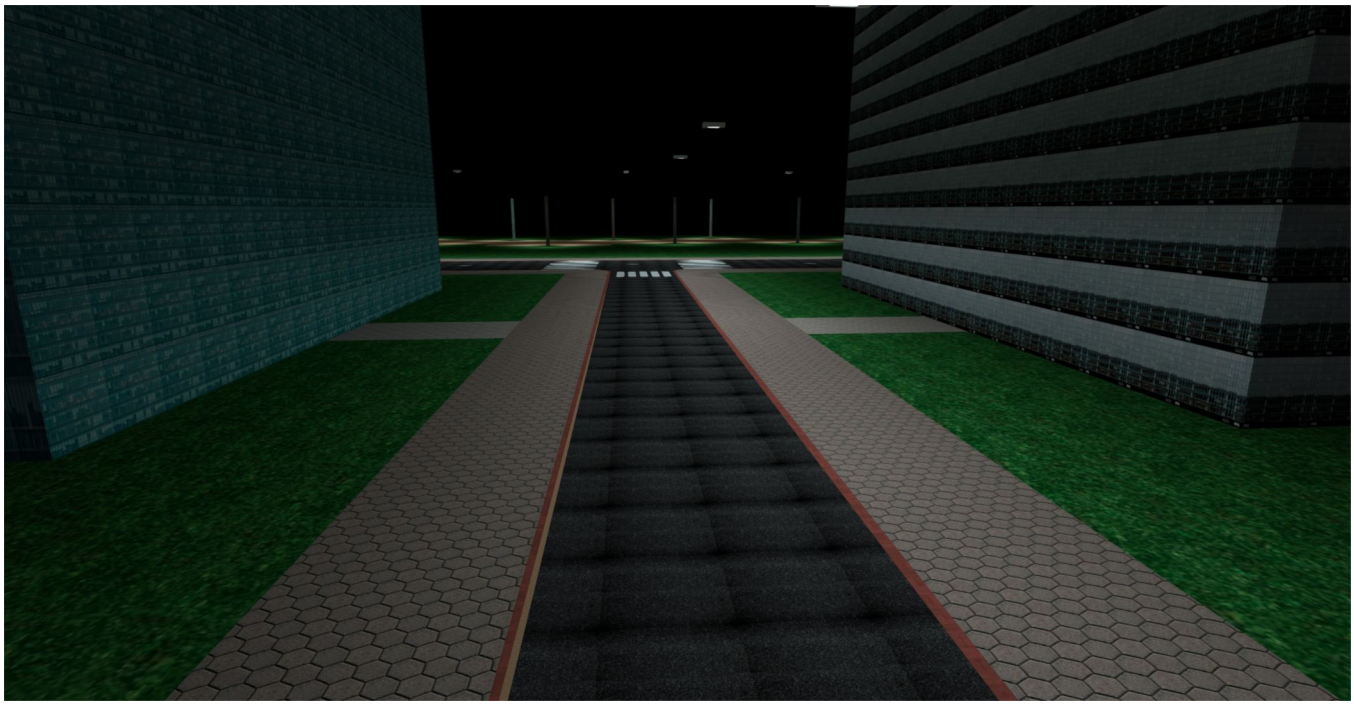
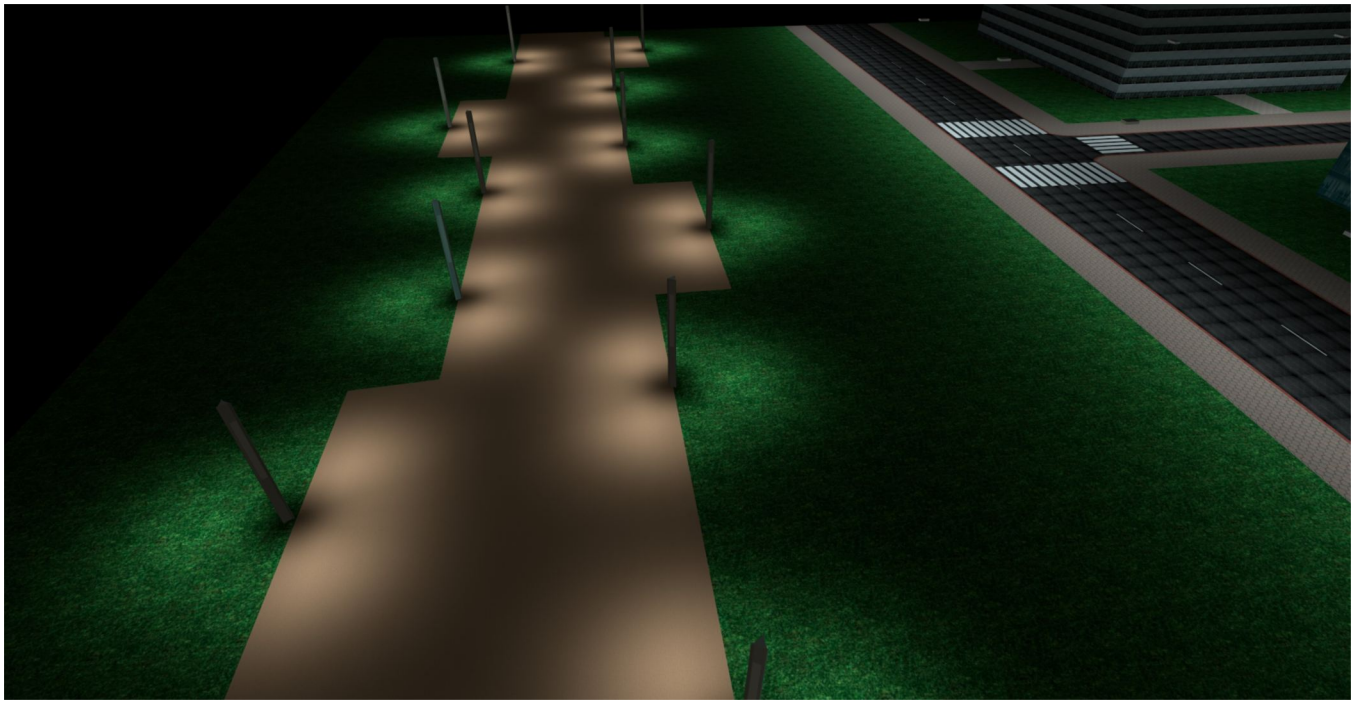


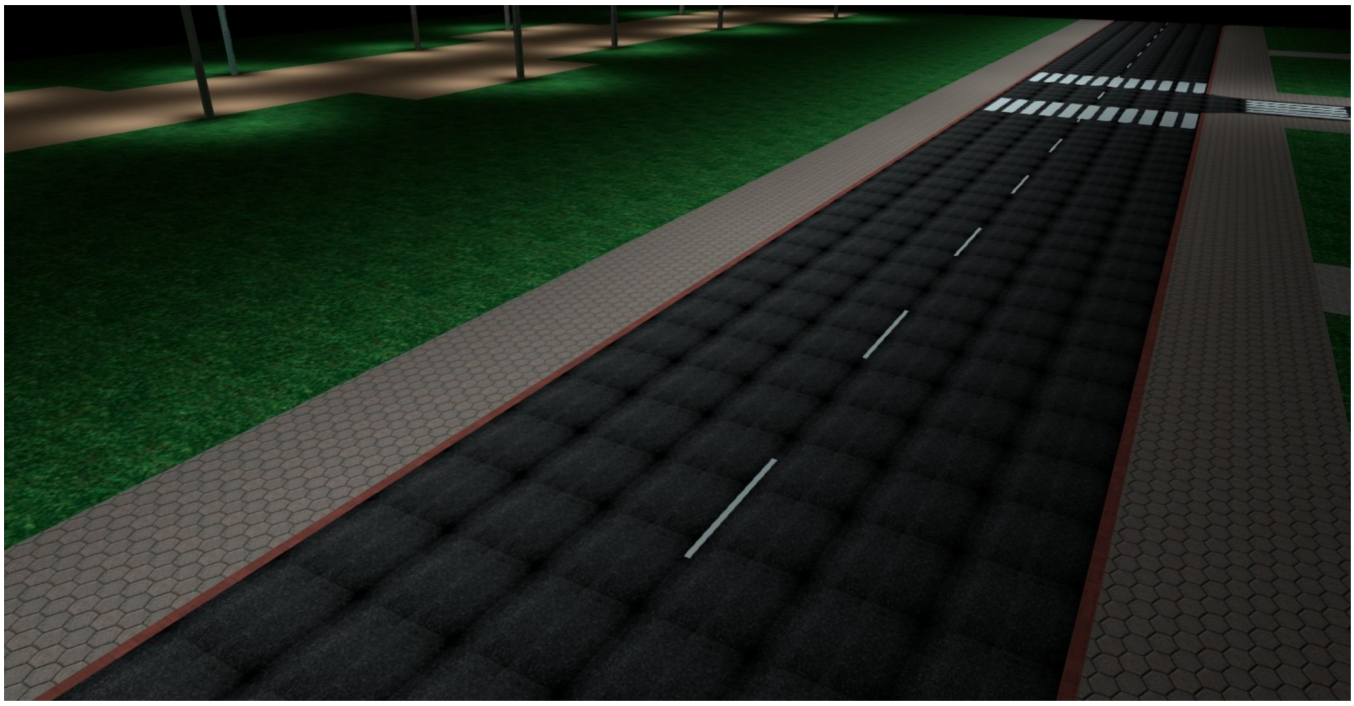


Images



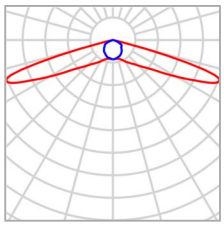


Images



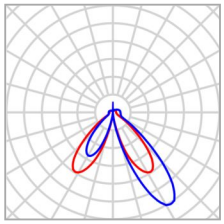
Luminaire list

Φ_{total}	P_{total}	Luminous efficacy
73722 lm	900.0 W	81.9 lm/W



pcs.	10	P	39.6 W
Manufacturer	ELEKTRO LUMEN	Φ_{Lamp}	5000 lm
Article No.	MARUT M G2 M16 5k0 722 B164	$\Phi_{Luminaire}$	4551 lm
Article name	MARUT M G2 M16 5k0 722 B164	η	91.03 %
Fitting	1x LED	Luminous efficacy	114.9 lm/W
		CCT	2200 K
		CRI	70

Luminaire list

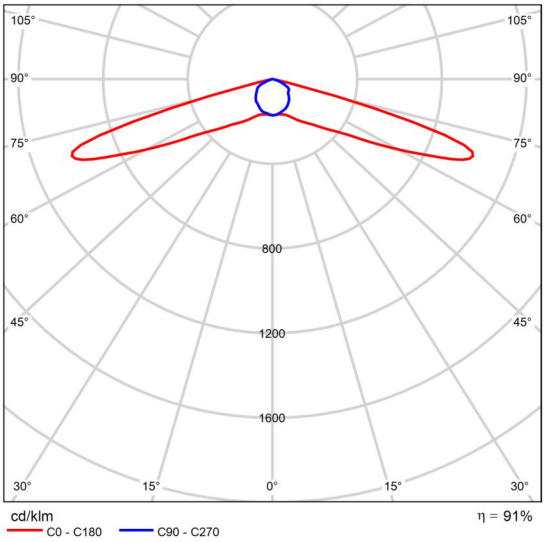


pcs.	12	P	42.0 W
Manufacturer	WE-EF	Φ_{Lamp}	5890 lm
Article No.	116-0207	$\Phi_{Luminaire}$	2351 lm
Article name	LTP444 IP44:LED-FT-37W/3K	η	39.92 %
Fitting	1x LED-FT-37W/830 - 3000K	Luminous efficacy	56.0 lm/W
		CCT	3000 K
		CRI	80

Product data sheet

ELEKTRO LUMEN - MARUT M G2 M16 5k0 722 B164

Article No.	MARUT M G2 M16 5k0 722 B164
P	39.6 W
Φ_{Lamp}	5000 lm
$\Phi_{Luminaire}$	4551 lm
η	91.03 %
Luminous efficacy	114.9 lm/W
CCT	2200 K
CRI	70



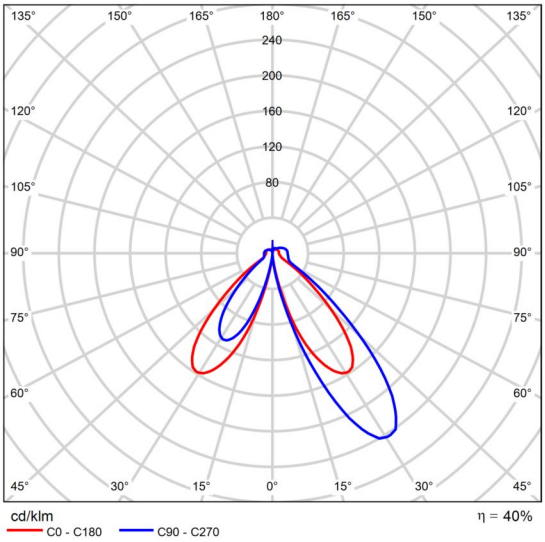
Polar LDC

Our best-selling line of compact LED luminaries for road and motorway lighting.

Product data sheet

WE-EF - LTP444 IP44:LED-FT-37W/3K

Article No.	116-0207
P	42.0 W
Φ_{Lamp}	5890 lm
$\Phi_{Luminaire}$	2351 lm
η	39.92 %
Luminous efficacy	56.0 lm/W
CCT	3000 K
CRI	80

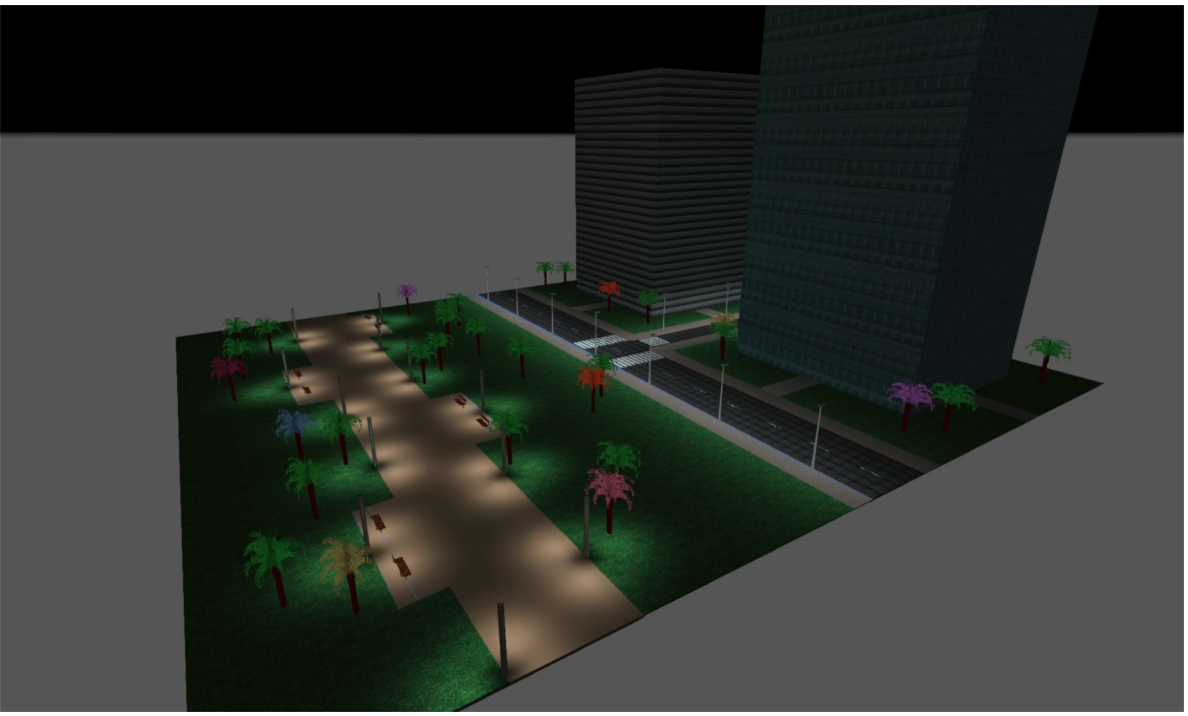


Polar LDC

IP44, Class I. IK10. Marine-grade, all aluminium construction. 5CE superior corrosion protection including PCS hardware. Silicone rubber gaskets. Prismatic polycarbonate lenses, 3 x 120° offset. Anodised aluminium reflector. Factory installed LED circuit board. LED boards can be easily removed for upgrading.
Service door with fused cable connecting box.

Trilight Column
Three sided light output
HIT 35 W

IP44. Marine-grade all aluminium construction. 5CE superior corrosion protection including PCS hardware. Powdercoat finish in black RAL 9004, grey aluminium RAL 9007 or white RAL 9016. Silicone rubber gaskets. Prismatic polycarbonate lenses. Service door with fused cable connecting box. Integral electrical control gear.



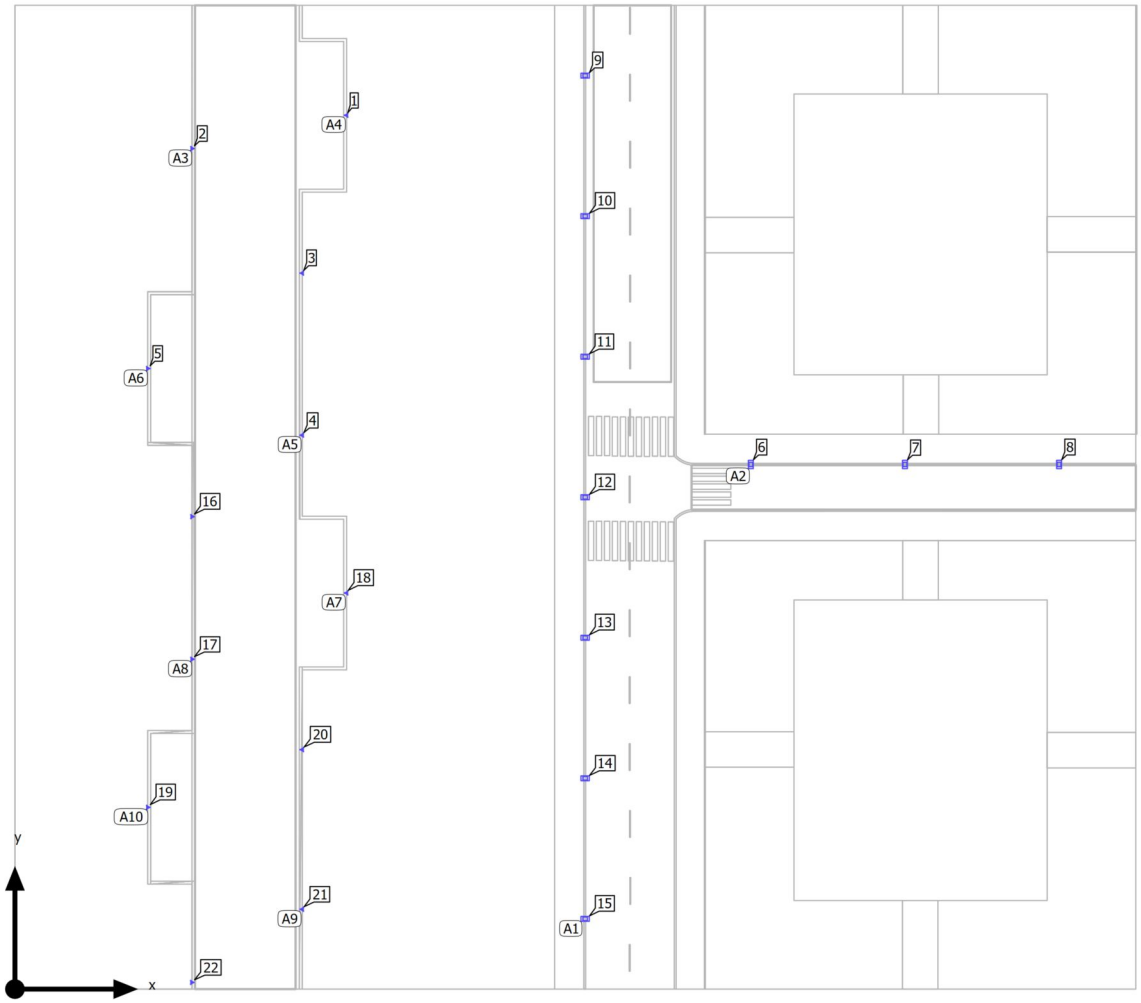
SITE

Description

The site comprises a mixed-use outdoor area including a primary roadway, pedestrian walkways, building surroundings, and landscaped zones. The lighting design considers functional requirements such as roadway illumination, pedestrian safety, and spatial visibility. The layout has been modeled in DIALux with accurate geometry to ensure reliable calculation of illuminance levels, uniformity, and glare control.

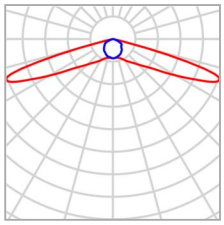
SITE

Luminaire layout plan



SITE

Luminaire layout plan



Manufacturer	ELEKTRO LUMEN	P	39.6 W
Article No.	MARUT M G2 M16 5k0 722 B164	$\Phi_{Luminaire}$	4551 lm
Article name	MARUT M G2 M16 5k0 722 B164		
Fitting	1x LED		

7 x ELEKTRO LUMEN MARUT M G2 M16 5k0 722 B164

Type	Line arrangement
1st luminaire (X/Y/Z)	38.602 m / 61.843 m / 5.100 m
X-direction	7 pcs., Centre - centre, 9.514 m
Arrangement	A1

X	Y	Mounting height	Luminaire
38.602 m	61.843 m	5.100 m	9
38.602 m	52.329 m	5.100 m	10
38.602 m	42.814 m	5.100 m	11
38.602 m	33.300 m	5.100 m	12
38.602 m	23.786 m	5.100 m	13
38.602 m	14.271 m	5.100 m	14
38.602 m	4.757 m	5.100 m	15

3 x ELEKTRO LUMEN MARUT M G2 M16 5k0 722 B164

Type	Line arrangement
1st luminaire (X/Y/Z)	49.820 m / 35.518 m / 5.100 m
X-direction	3 pcs., Centre - centre, 10.435 m

X	Y	Mounting height	Luminaire
49.820 m	35.518 m	5.100 m	6
60.255 m	35.518 m	5.100 m	7

SITE

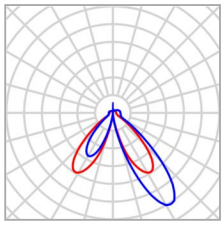
Luminaire layout plan

Arrangement	A2

X	Y	Mounting height	Luminaire
70.690 m	35.518 m	5.100 m	8

SITE

Luminaire layout plan



Manufacturer	WE-EF	P	42.0 W
Article No.	116-0207	$\Phi_{Luminaire}$	2351 lm
Article name	LTP444 IP44:LED-FT-37W/3K		
Fitting	1x LED-FT-37W/830 - 3000K		

1 x WE-EF LTP444 IP44:LED-FT-37W/3K

Type	Line arrangement
1st luminaire (X/Y/Z)	11.989 m / 56.906 m / 0.100 m
X-direction	1 pcs., Centre - centre, 19.387 m
Arrangement	A3

X	Y	Mounting height	Luminaire
11.989 m	56.906 m	0.100 m	2

1 x WE-EF LTP444 IP44:LED-FT-37W/3K

Type	Line arrangement
1st luminaire (X/Y/Z)	22.453 m / 59.157 m / 0.100 m
X-direction	1 pcs., Centre - centre, 10.400 m
Arrangement	A4

X	Y	Mounting height	Luminaire
22.453 m	59.157 m	0.100 m	1

2 x WE-EF LTP444 IP44:LED-FT-37W/3K

SITE

Luminaire layout plan

Type	Line arrangement
1st luminaire (X/Y/Z)	19.453 m / 48.472 m / 0.100 m
X-direction	2 pcs., Centre - centre, 10.971 m
Arrangement	A5

X	Y	Mounting height	Luminaire
19.453 m	48.472 m	0.100 m	3
19.453 m	37.501 m	0.100 m	4

1 x WE-EF LTP444 IP44:LED-FT-37W/3K

Type	Line arrangement
1st luminaire (X/Y/Z)	8.991 m / 42.013 m / 0.100 m
X-direction	1 pcs., Centre - centre, 10.400 m
Arrangement	A6

X	Y	Mounting height	Luminaire
8.991 m	42.013 m	0.100 m	5

1 x WE-EF LTP444 IP44:LED-FT-37W/3K

Type	Line arrangement
1st luminaire (X/Y/Z)	22.453 m / 26.815 m / 0.100 m
X-direction	1 pcs., Centre - centre, 10.400 m
Arrangement	A7

X	Y	Mounting height	Luminaire
22.453 m	26.815 m	0.100 m	18

2 x WE-EF LTP444 IP44:LED-FT-37W/3K

Type	Line arrangement
1st luminaire (X/Y/Z)	11.988 m / 31.986 m / 0.100 m
X-direction	2 pcs., Centre - centre, 9.653 m
Arrangement	A8

X	Y	Mounting height	Luminaire
11.988 m	31.986 m	0.100 m	16
11.990 m	22.334 m	0.100 m	17

SITE

Luminaire layout plan

2 x WE-EF LTP444 IP44:LED-FT-37W/3K

Type	Line arrangement
1st luminaire (X/Y/Z)	19.453 m / 16.211 m / 0.100 m
X-direction	2 pcs., Centre - centre, 10.808 m
Arrangement	A9

X	Y	Mounting height	Luminaire
19.453 m	16.211 m	0.100 m	20
19.453 m	5.404 m	0.100 m	21

1 x WE-EF LTP444 IP44:LED-FT-37W/3K

Type	Line arrangement
1st luminaire (X/Y/Z)	8.991 m / 12.307 m / 0.100 m
X-direction	1 pcs., Centre - centre, 10.400 m
Arrangement	A10

X	Y	Mounting height	Luminaire
8.991 m	12.307 m	0.100 m	19

Individual luminaires

X	Y	Mounting height	Luminaire
11.976 m	0.454 m	0.100 m	22

SITE

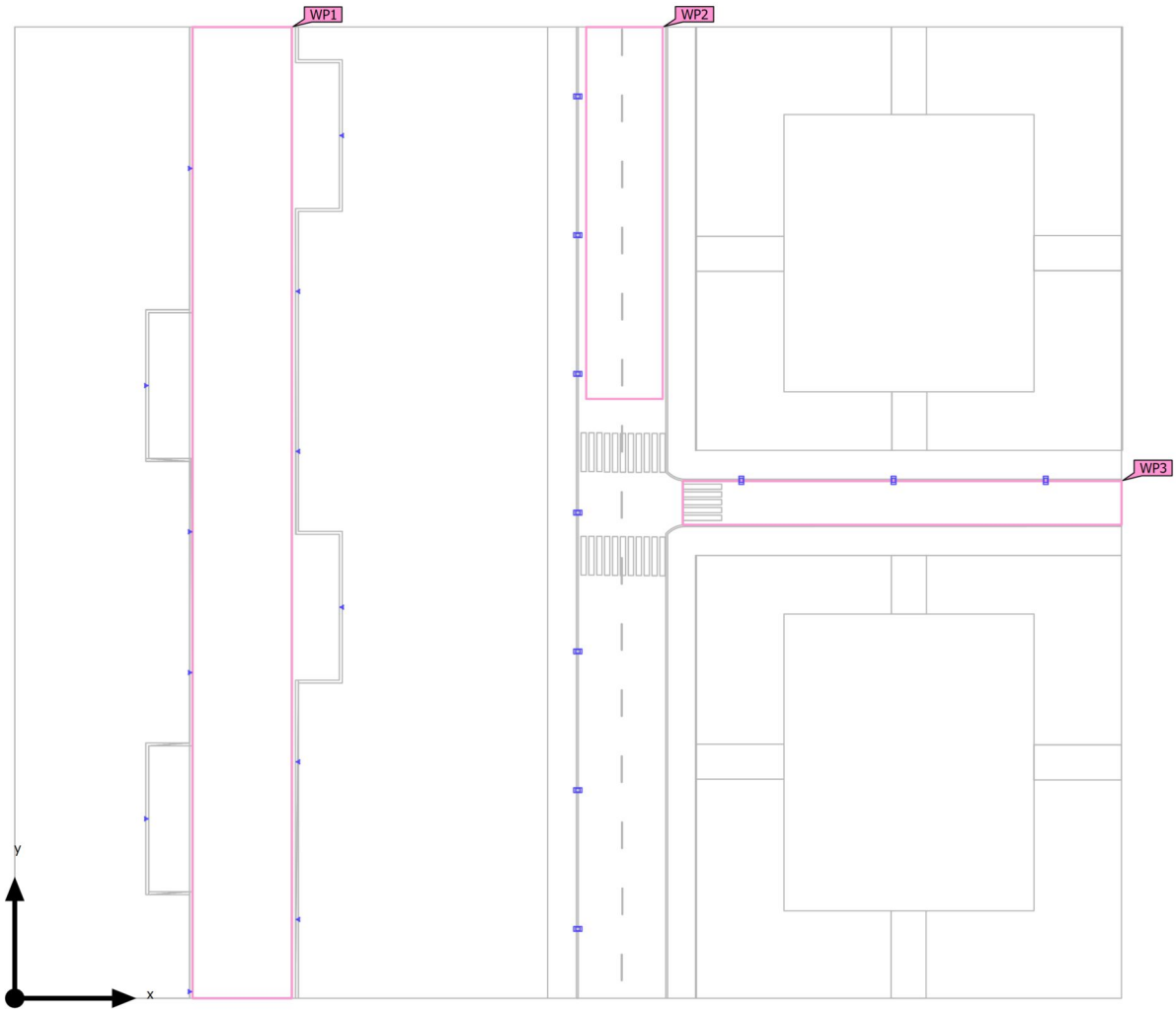
Luminaire list

Φ_{total}	P_{total}	Luminous efficacy
73722 lm	900.0 W	81.9 lm/W

pcs.	Manufacturer	Article No.	Article name	P	Φ	Luminous efficacy
10	ELEKTRO LUMEN	MARUT M G2 M16 5k0 722 B164	MARUT M G2 M16 5k0 722 B164	39.6 W	4551 lm	114.9 lm/W
12	WE-EF	116-0207	LTP444 IP44:LED-FT-37W/3K	42.0 W	2351 lm	56.0 lm/W

SITE (Light scene 1)

Calculation objects



SITE (Light scene 1)

Calculation objects

Working planes

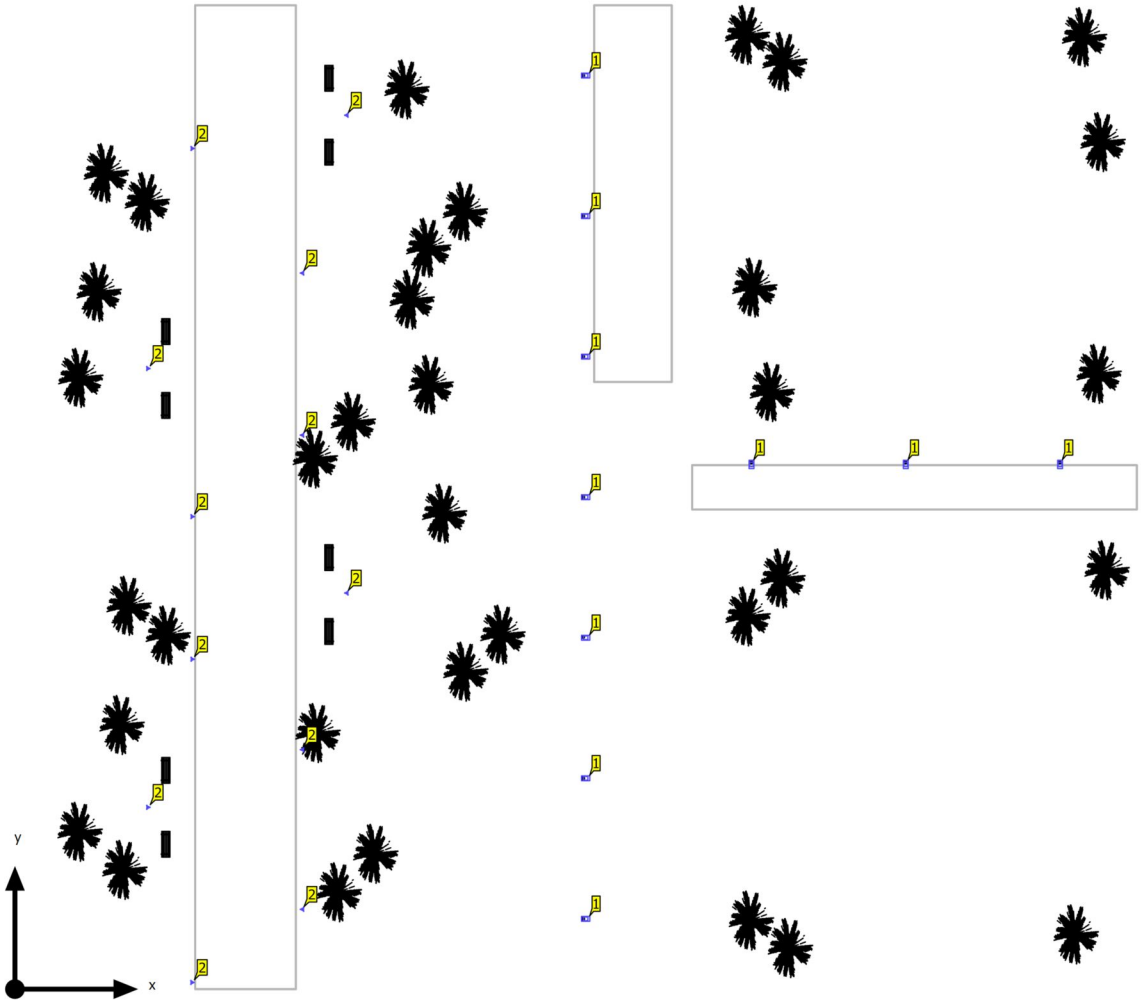
Properties	Ē (Target)	E_{min}	E_{max}	U_o (g_1) (Target)	g_2	Index
Working plane (Main Road) Perpendicular illuminance (adaptive) Height (Space): 0.000 m, Height (Site): 0.100 m, Wall zone: 0.000 m	23.8 lx (≥ 20.0 lx) ✓	5.29 lx	50.8 lx	0.22 (≥ 0.22) ✓	0.10	WP2
Working plane (Mid Road) Perpendicular illuminance (adaptive) Height (Space): 0.000 m, Height (Site): 0.100 m, Wall zone: 0.000 m	31.0 lx (≥ 20.0 lx) ✓	12.6 lx	48.8 lx	0.41 (≥ 0.40) ✓	0.26	WP3
Working plane (Pedestrian Space) Perpendicular illuminance (adaptive) Height (Space): 0.000 m, Height (Site): 0.100 m, Wall zone: 0.000 m	13.7 lx (≥ 5.00 lx) ✓	0.39 lx	59.1 lx	0.028 (≥ 0.025) ✓	0.007	WP1

Utilisation profile: DIALux presetting (5.1.4 Standard (outdoor transportation area))

Notes on planning:
The results were calculated without consideration of objects and furniture. No results were determined on their surfaces.

SITE

Control group CG 1



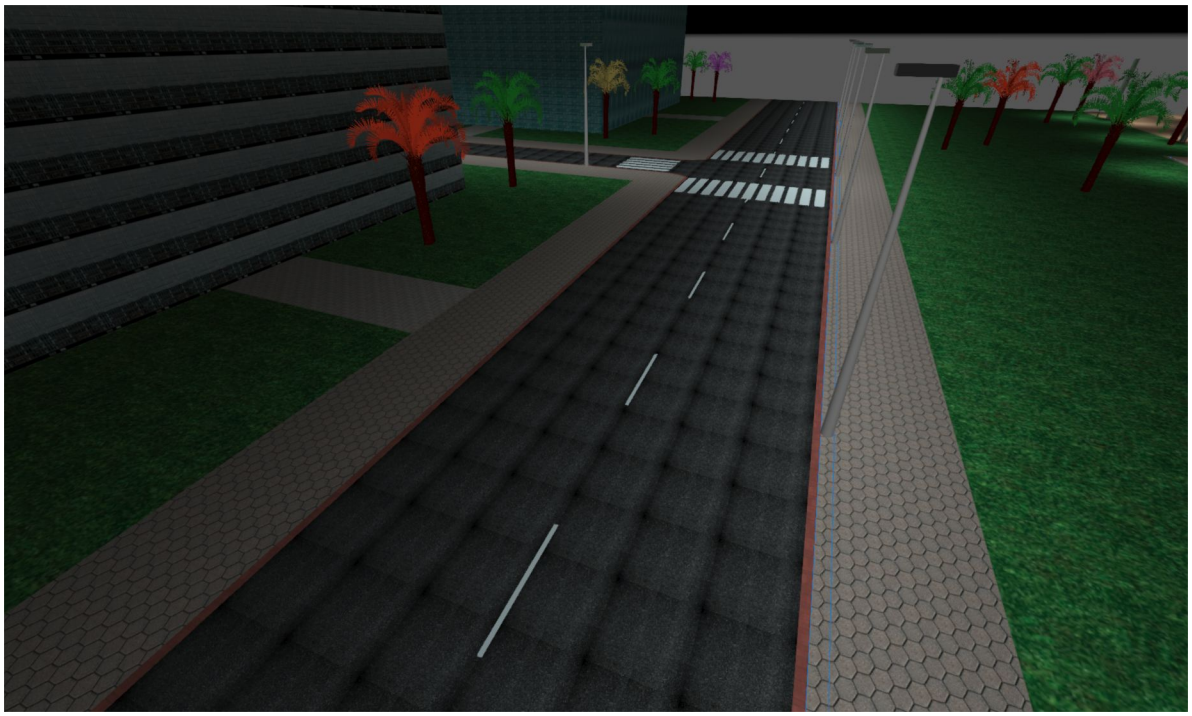
SITE

Control groups

Control group	CG 1
Light scene 1	100

Dimming values [%]

pcs.	Manufacturer	Article No.	Article name	P	Index
10	ELEKTRO LUMEN	MARUT M G2 M16 5k0 722 B164	MARUT M G2 M16 5k0 722 B164	39.6 W	1
12	WE-EF	116-0207	LTP444 IP44:LED-FT-37W/3K	42.0 W	2



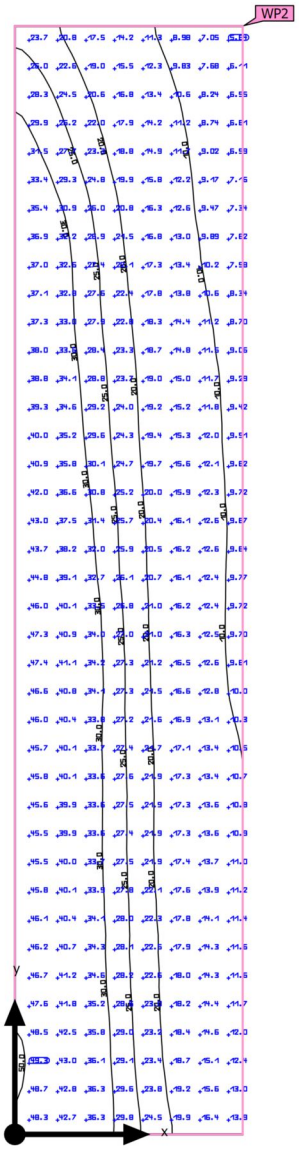
Main Road

Description

The roadway lighting system is designed in accordance with standard road lighting criteria, focusing on achieving required average illuminance, uniformity ratio, and glare control. Pole height, spacing, and luminaire distribution have been optimized to ensure consistent light coverage and safe driving conditions during nighttime operations.

Main Road (Light scene 1)

Summary



Ground area	133.75 m^2	Height Working plane	0.000 m
Maintenance factor	0.80 (fixed)	Wall zone Working plane	0.000 m

Main Road (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{perpendicular}$	23.8 lx	≥ 20.0 lx	✓	WP2
	U_o (g_1)	0.22	≥ 0.22	✓	WP2
Energy estimation(2)	Consumption	0.00 kWh/a	max. 4700 kWh/a	✓	
Space	Lighting power density	0.00 W/m²	–		
		0.00 W/m²/100 lx	–		

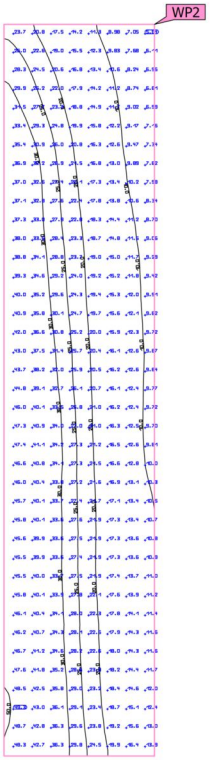
(1) Based on a rectangular space of 25.500 m x 5.245 m and SHR of 0.25.
(2) Calculated using DIN:18599-4.

Utilisation profile: General circulation areas at outdoor workplaces (5.1.3 Regular vehicle traffic (max. 40km/h))

Notes on planning:
The results were calculated without consideration of objects and furniture. No results were determined on their surfaces.

Main Road (Light scene 1)

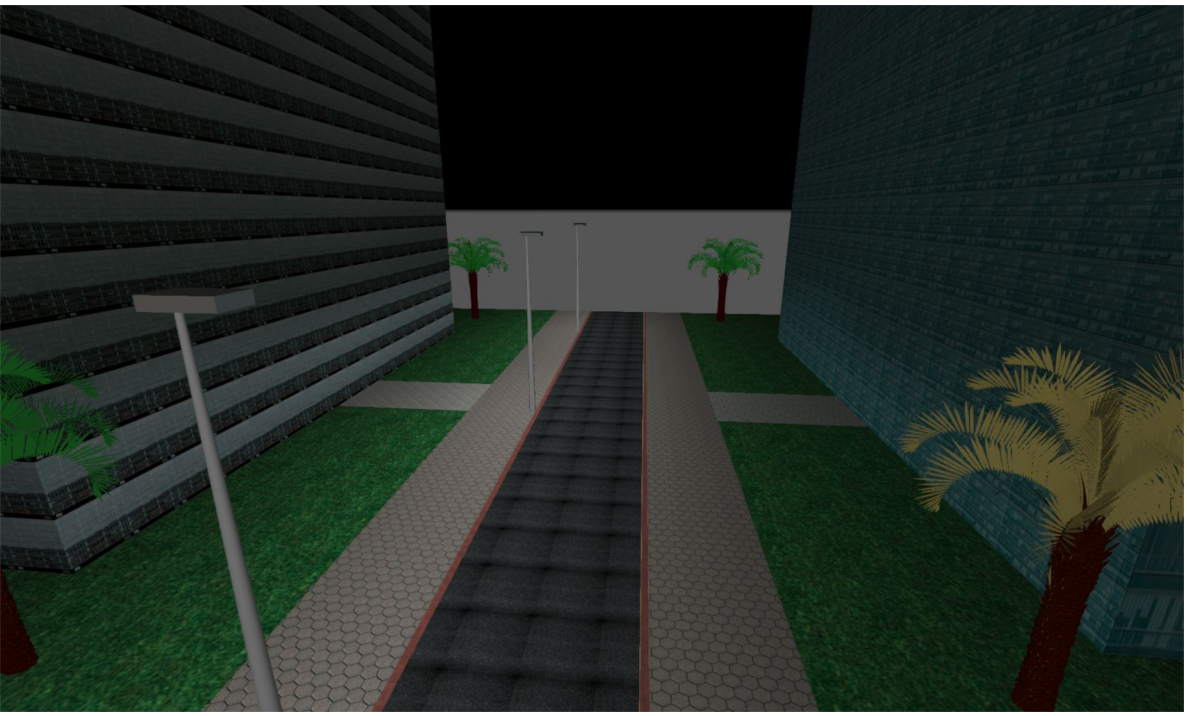
Working plane (Main Road)



Properties	Ē (Target)	E_{min}	E_{max}	U_o (g_1) (Target)	g_2	Index
Working plane (Main Road) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	23.8 lx (≥ 20.0 lx) ✓	5.29 lx	50.8 lx	0.22 (≥ 0.22) ✓	0.10	WP2

Utilisation profile: General circulation areas at outdoor workplaces (5.1.3 Regular vehicle traffic (max. 40km/h))

Notes on planning:
The results were calculated without consideration of objects and furniture. No results were determined on their surfaces.



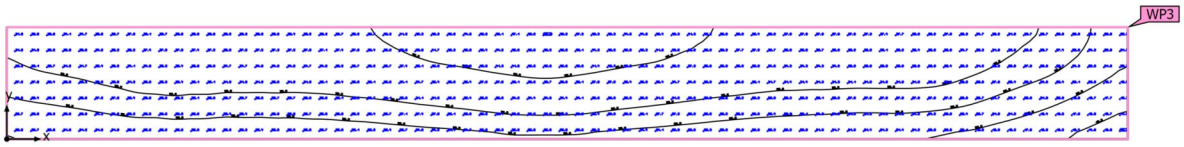
Mid Road

Description

The roadway lighting system is designed in accordance with standard road lighting criteria, focusing on achieving required average illuminance, uniformity ratio, and glare control. Pole height, spacing, and luminaire distribution have been optimized to ensure consistent light coverage and safe driving conditions during nighttime operations.

Mid Road (Light scene 1)

Summary



Ground area	90.28 m^2	Height Working plane	0.000 m
Maintenance factor	0.80 (fixed)	Wall zone Working plane	0.000 m

Mid Road (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{perpendicular}$	31.0 lx	≥ 20.0 lx	✓	WP3
	U_o (g_1)	0.41	≥ 0.40	✓	WP3
Energy estimation(2)	Consumption	0.00 kWh/a	max. 3200 kWh/a	✓	
Space	Lighting power density	0.00 W/m²	–		
		0.00 W/m²/100 lx	–		

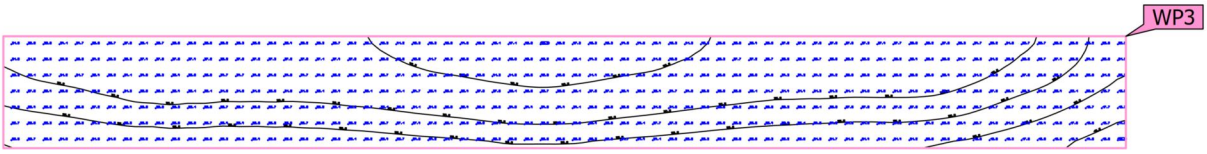
(1) Based on a rectangular space of 30.078 m x 3.001 m and SHR of 0.25.
(2) Calculated using DIN:18599-4.

Utilisation profile: General circulation areas at outdoor workplaces (5.1.3 Regular vehicle traffic (max. 40km/h))

Notes on planning:
The results were calculated without consideration of objects and furniture. No results were determined on their surfaces.

Mid Road (Light scene 1)

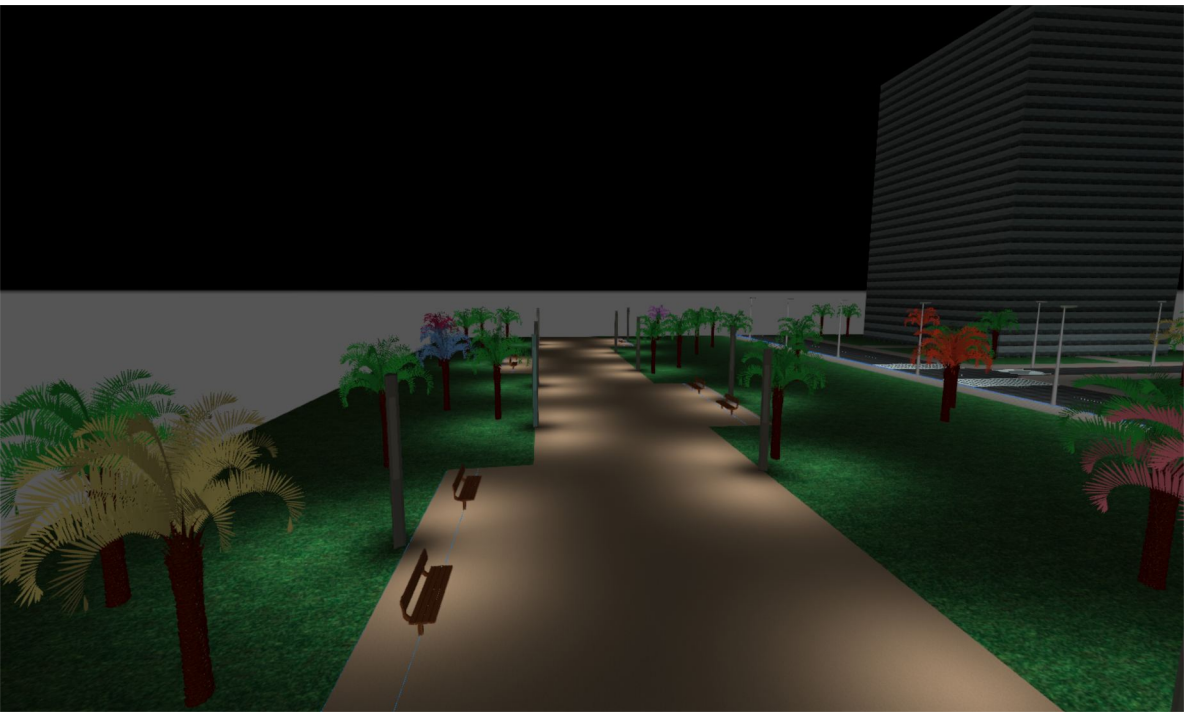
Working plane (Mid Road)



Properties	Ē (Target)	E_{min}	E_{max}	U_o (g_1) (Target)	g_2	Index
Working plane (Mid Road) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	31.0 lx (≥ 20.0 lx) ✓	12.6 lx	48.8 lx	0.41 (≥ 0.40) ✓	0.26	WP3

Utilisation profile: General circulation areas at outdoor workplaces (5.1.3 Regular vehicle traffic (max. 40km/h))

Notes on planning:
The results were calculated without consideration of objects and furniture. No results were determined on their surfaces.



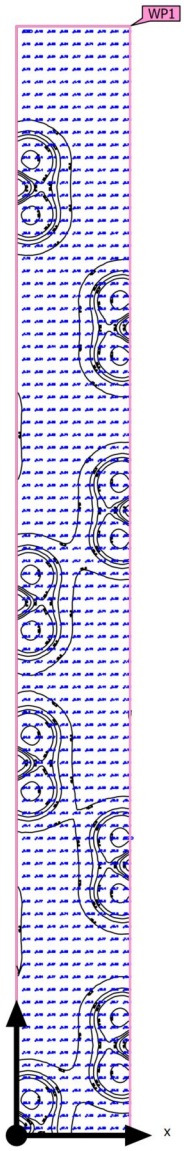
Pedestrian Space

Description

The pedestrian areas are designed to provide safe and comfortable movement during nighttime through adequate and uniform lighting. Luminaires are positioned to ensure clear visibility along walkways while minimizing glare and dark spots. The design enhances user safety, orientation, and overall visual comfort.

Pedestrian Space (Light scene 1)

Summary



Ground area	453.46 m^2	Height Working plane	0.000 m
Maintenance factor	0.80 (fixed)	Wall zone Working plane	0.000 m

Pedestrian Space (Light scene 1)

Summary

Results

	Symbol	Calculated	Target	Check	Index
Working plane	$\bar{E}_{perpendicular}$	13.7 lx	≥ 5.00 lx	✓	WP1
	U_o (g_1)	0.028	≥ 0.025	✓	WP1
Energy estimation(2)	Consumption	0.00 kWh/a	max. 15900 kWh/a	✓	
Space	Lighting power density	0.00 W/m²	–		
		0.00 W/m²/100 lx	–		

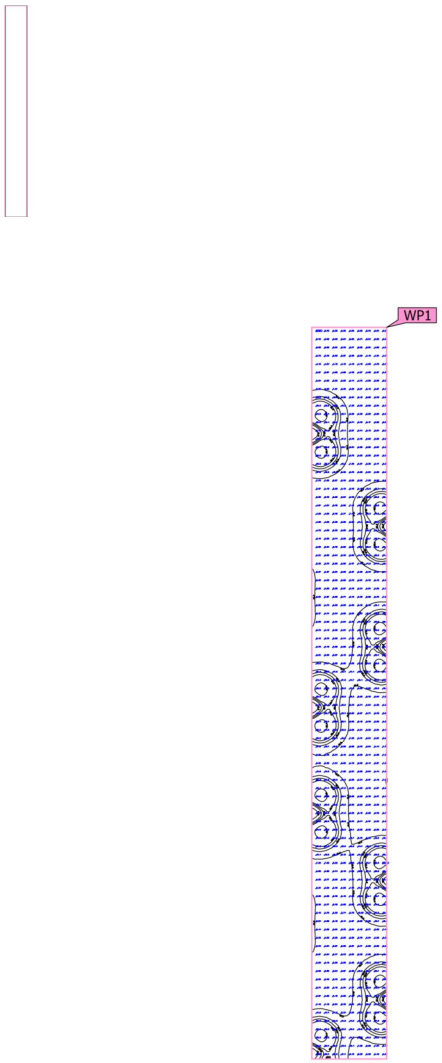
(1) Based on a rectangular space of 66.600 m x 6.809 m and SHR of 0.25.
(2) Calculated using DIN:18599-4.

Utilisation profile: General circulation areas at outdoor workplaces (5.1.1 Walkways exclusively for pedestrians)

Notes on planning:
The results were calculated without consideration of objects and furniture. No results were determined on their surfaces.

Pedestrian Space (Light scene 1)

Working plane (Pedestrian Space)



Properties	Ē (Target)	E_{min}	E_{max}	U_o (g_1) (Target)	g_2	Index
Working plane (Pedestrian Space) Perpendicular illuminance (adaptive) Height: 0.000 m, Wall zone: 0.000 m	13.7 lx (≥ 5.00 lx) ✓	0.39 lx	59.1 lx	0.028 (≥ 0.025) ✓	0.007	WP1

Utilisation profile: General circulation areas at outdoor workplaces (5.1.1 Walkways exclusively for pedestrians)

Notes on planning:
The results were calculated without consideration of objects and furniture. No results were determined on their surfaces.

Glossary

A

A	Formula symbol for a surface in the geometry

B

Background area	The background area borders the direct ambient area according to DIN EN 12464-1 and reaches up to the borders of the room. In larger rooms, the background area is at least 3 m wide. It is located horizontally at floor level.

C

CCT	(Engl. correlated color temperature) Body temperature of a thermal radiator which serves to describe its light color. Unit: Kelvin [K]. The lesser the numerical value the redder; the greater the numerical value the bluer the light colour. The color temperature of gas-discharge lamps and semi-conductors are termed "correlated color temperature" in contrast to the color temperature of thermal radiators. Allocation of the light colors to the color temperature ranges acc. to EN 12464-1: Light color - color temperature [K] warm white (ww) < 3,300 K neutral white (nw) ≥ 3,300 – 5,300 K daylight white (dw) > 5,300 K
Clearance height	The designation for the distance between upper edge of the floor and bottom edge of the ceiling (in the completely furnished status of room).
Control group	A group of luminaires that are dimmed and controlled together. For each lighting scene, a control group provides its own dimming value. All luminaires within a control group share this dimming value. The control groups with their luminaires are automatically determined by DIALux on the basis of the created light scenes and their luminaire groups.
CRI	(Engl. color rendering index) Designation for the color rendering index of a luminaire or a lamp acc. to DIN 6169: 1976 or CIE 13.3: 1995. The general color rendering index Ra (or CRI) is a dimensionless figure that describes the quality of a white light source in regards to its similarity with the remission spectra of defined 8 test colors (see DIN 6169 or CIE 1974) to a reference light source.

Glossary

D

Daylight autonomy	Describes what percentage of the daily working time the required illuminance is met by daylight. The nominal illuminance is used from the room profile, unlike described in EN 17037. The calculation is not done in the centre of the room but at the placed sensor measuring point. A room is considered sufficiently supplied with daylight if it achieves at least 50% daylight autonomy.
Daylight factor	Ratio of the illuminance achieved solely by daylight incidence at a point in the inside to the horizontal illuminance in the outer area under an unobstructed sky. Formula symbol: D (Engl. daylight factor) Unit: %
Daylight quotient effective area	A calculation surface within which the daylight quotient is calculated.

E

Energy evaluation	Based on an hourly calculation procedure for daylight in indoor spaces, considering the project geometry and any existing daylight control systems. Orientation and location of the project are also considered. The calculation uses the specified system power of the luminaires to determine the energy demand. A linear relationship between power and luminous flux in the dimmed state is assumed for daylight-controlled luminaires. Times of use and nominal illuminance are determined from the usage profiles of the spaces. Switched-on luminaires that are explicitly excluded from control also consider the specified times-of-use. The daylight control systems use a simplified control logic that closes them at an outdoor horizontal illuminance of 27,500lx. The calendar year 2022 is used as a reference only. It is not a simulation of this year. The reference year is only used to assign the days of the week to the calculated results. The changeover to summer time is not considered. The reference sky type used is the average sky described in CIE 110 without direct sunlight. The method was developed together with the Fraunhofer Institute for Building Physics and is available for review by the Joint Working Group 1 ISO TC 274 as an extension of the previous annual regression-based method.
Environmental zones	The assessment of intrusive light and light immission depends on the environment of the lighting installation. Depending on the standard, 4-6 different zones are defined, ranging from highly protected areas in natural settings to urban areas, commercial zones, and industrial zones.
Eta (η)	(light output ratio) The light output ratio describes what percentage of the luminous flux of a free radiating lamp (or LED module) is emitted by the luminaire when installed. Unit: %

Glossary

G

g_1	Often also U_o (Engl. overall uniformity) Designates the overall uniformity of the illuminance on a surface. It is the quotient from E_{min} to $\bar{E}$ and is required, for instance, in standards for illumination of workstations.
g_2	Actually it designates the "non-uniformity" of the illuminance on a surface. It is the quotient of E_{min} to E_{max} and is generally only relevant for certifying the emergency lighting acc. to EN 1838.

I

Illuminance	Describes the ratio of the luminous flux that strikes a certain surface to the size of this surface (lm/m^2 = lx). The illuminance is not tied to an object surface. It can be determined anywhere in space (inside or outside). The illuminance is not a product feature because it is a recipient value. Luxometers are used for measuring. Unit: Lux Abbreviation: lx Formula symbol: E
Illuminance, adaptive	For the determining of the middle adaptive illuminance on a surface, this is rastered "adaptively". In the area of large illuminance differences within the surface, the raster is subdivided finer; within lesser differences, a rougher classification is made.
Illuminance, horizontal	Illuminance that is calculated or measured on a horizontal (level) surface (this can be for example a table top or the floor). The horizontal illuminance is usually identified by the formula letter E_h.
Illuminance, perpendicular	Illuminance that is calculated or measured plumb-vertical to a surface. This needs to be taken into account for tilted surfaces. If the surface is horizontal or vertical, then there is no difference between the perpendicular and the horizontal or vertical illuminance.
Illuminance, vertical	Illuminance that is calculated or measured on a vertical surface (this can be for example the front of some shelves). The vertical illuminance is usually identified by the formula letter E_v.

K

k_S	The glare effect of a light source can be described by the glare metric k_S. It relates the solid angle of the glaring light source as seen from the point of immission, the ambient luminance, and the maximum allowable luminance.

Glossary

L

LENI	(Engl. lighting energy numeric indicator) Lighting energy numeric indicator acc. to EN 15193 Unit: kWh/(m^2 * a)
LLMF	(Engl. lamp lumen maintenance factor)/acc. to CIE 97: 2005 Lamp flux maintenance factor that takes the luminous flux reduction into account of a luminaire or an LED module in the course of the operating time. The lamp flux maintenance factor is specified as a decimal digit and can have a maximum value of 1 (no luminous flux reduction existing).
LMF	(Engl. luminaire maintenance factor)/acc. to CIE 97: 2005 Luminaire maintenance factor that takes the soiling into account of the luminaire in the course of the operating time. The luminaire maintenance factor is specified as a decimal digit and can have a maximum value of 1 (no soiling existing).
LSF	(Engl. lamp survival factor)/acc. to CIE 97: 2005 Lamp survival factor that takes the total failure into account of a luminaire in the course of the operating time. The lamp survival factor is specified as a decimal digit and can have a maximum value of 1 (no failures existing within the time concerned or prompt replacement after the failure).
Luminance	Dimension for the "brightness impression" that the human eye has of a surface. The surface itself can emit light thereby or light striking it can be reflected (emitter value). It is the only photometric value that the human eye can perceive. Unit: Candela per square metre Abbreviation: cd/m^2 Formula symbol: L
Luminous efficacy	Ratio of the emitted luminous flux Φ [lm] to the absorbed electrical power P [W] Unit: lm/W. This ratio can be formed for the lamp or LED module (lamp or module light output), the lamp or module with control gear (system light output) and the complete luminaire (luminaire light output).
Luminous flux	Dimension for the total light output that is emitted from one light source in all directions. It is thus an "emitter value" that specifies the entire emitting output. The luminous flux of a light source can only be determined in a laboratory. A difference is made between the lamp or LED module luminous flux and the luminaire luminous flux. Unit: Lumen Abbreviation: lm Formula symbol: Φ

Glossary

Luminous intensity	Describes the intensity of the light in a certain direction (emitter value). The luminous intensity is a matter of the luminous flux Φ that is emitted in a certain spherical angle Ω. The radiation characteristics of a light source are presented graphically in a light distribution curve (LDC). The luminous intensity is an SI base unit. Unit: Candela Abbreviation: cd Formula symbol: I

M

Maintenance factor	See MF
MF	(Engl. maintenance factor)/acc. to CIE 97: 2005 Maintenance factor as decimal number between 0 and 1 that describes the ratio of the new value of a photometric planning parameter (e.g. of the illuminance) to a maintenance value after a certain time. The maintenance factor takes into account the soiling of luminaires and rooms as well as the luminous flux reduction and the failure of light sources. The maintenance factor is taken into account either overall or determined in detail acc. to CIE 97: 2005 by the formula RMF x LMF x LLMF x LSF.

O

Obtrusive light/Light immission	To protect the nocturnal environment and minimize problems for humans, flora, and fauna, it is necessary to limit obtrusive light (also known as light pollution), which can cause serious physiological and ecological issues for individuals and the environment. Light immission refers to the disturbing influence of emitted light from artificial light sources.
Operating times	The assessment of obtrusive light and light immission depends on the operating times of the lighting installation. Depending on the standard, 1-3 different operating times are specified. In the absence of specific details, an operating time between 06:00 and 22:00 can be assumed.

P

P	(Engl. power) Electric power consumption Unit: watt Abbreviation: W

Glossary

R

$R_{(UG)}$ max	Measure of the psychological glare in indoor spaces. In addition to the luminance of luminaires, the level of the $R_{(UG)}$ value also depends on the observer position, the viewing direction and the ambient luminance. The calculation is made according to the table method, see CIE 117. Among other things, EN 12464-1:2021 specifies maximum permissible $R_{(UG)}$- values $R_{(UGL)}$ for various indoor workplaces.
R_{DLO}	The ratio of the luminous flux emitted below the horizontal plane to the total lamp luminous flux of a luminaire or lighting installation in its operational position.
R_G	The glare directly caused by luminaires of an outdoor lighting installation is determined using the CIE Glare Rating (RG) method. To calculate this, the equivalent veiling luminance of the surroundings is needed. There are four options for determining this: • An exact calculation according to CIE 112, based on the scene area. • A simplified method according to EN 12464-2, based on the scene area. • Using a custom calculation area to determine the equivalent veiling luminance. • Specifying a fixed value for easy comparability.
R_{UF}	upward flux ratio The ratio of the luminous flux emitted directly or reflected above the horizontal plane to the luminous flux that cannot be avoided under ideal conditions to achieve the illuminance level on a deliberately illuminated area.
R_{UL}	upward light ratio The ratio of the luminous flux emitted above the horizontal plane to the luminous flux of a luminaire or lighting installation in its operational position. The luminaire efficiency is considered in this calculation.
R_{ULO}	upward light output ratio The ratio of the luminous flux emitted above the horizontal plane to the total lamp luminous flux of a luminaire or lighting installation in its operational position.
Reflection factor	The reflection factor of a surface describes how much of the striking light is reflected back. The reflection factor is defined by the color of the surface.
RMF	(Engl. room maintenance factor)/acc. to CIE 97: 2005 Room maintenance factor that takes the soiling into account of the space encompassing surfaces in the course of the operating time. The room maintenance factor is specified as a decimal digit and can have a maximum value of 1 (no soiling existing).
RUG (max)	(unified glare rating) Measure for the psychological glare effect in interiors. In addition to luminaire luminance, the RUG value also depends on the position of the observer, the viewing direction and the ambient luminance. Among other things, EN 12464-1 specifies maximum permissible RUG values for various indoor workplaces.

Glossary

RUG observer	Calculation point in the room, for the DIALux the RUG value is determined. The location and height of the calculation point should correspond to the typical observer position (position and eye level of the user).

S

Surrounding area	The ambient area directly borders the area of the visual task and should be planned with a width of at least 0.5 m according to DIN EN 12464-1. It is at the same height as the area of the visual task.

V

Visual task area	The area that is needed for carrying out the visual task in accordance with DIN EN 12464-1. The height corresponds with the height at which the visual task is executed.

W

Wall zone	Circumferential area between working plane and walls which is not taken into account for the calculation.
Working plane	Virtual measuring or calculation surface at the height of the visual task that generally follows the room geometry. The working plane may also feature a wall zone.